Assignment8

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Introduction In this project, I analyzed forest fire incidents in Turkey between 2000 and 2021 using NASA FIRMS satellite data. The main objective is to visualize the spatial distribution of fires across provinces and understand the seasonal and annual trends. This analysis aims to highlight high-risk regions and temporal anomalies, such as the massive wildfires of 2021.

#1. Spatial Distribution: Fires by Province To identify the most vulnerable regions, I aggregated the fire incidents by province boundaries. Similar to demographic density mapping, this visualization highlights provinces with the highest frequency of fires.

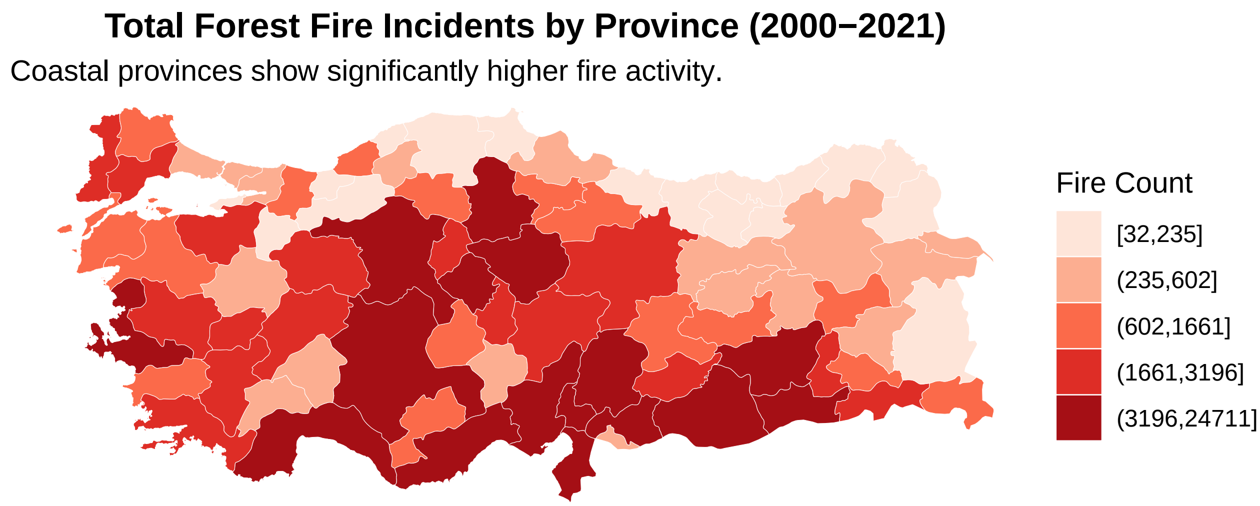
```
province_stats <- fire_joined %>%
  group_by(name_tr) %>%
  summarise(count = n()) %>%
  st_drop_geometry()


map_data <- left_join(turkey_map, province_stats, by = "name_tr")


map_data$count[is.na(map_data$count)] <- 0


map_data$count_cut <- cut_number(map_data$count, n = 5, dig.lab = 50)


ggplot(map_data) +
  geom_sf(aes(fill = count_cut), color = "white", size = 0.1) +
  scale_fill_brewer(palette = "Reds", name = "Fire Count") +
  labs(title = "Total Forest Fire Incidents by Province (2000-2021)",
       subtitle = "Coastal provinces show significantly higher fire activity.") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
        legend.position = "right")
```

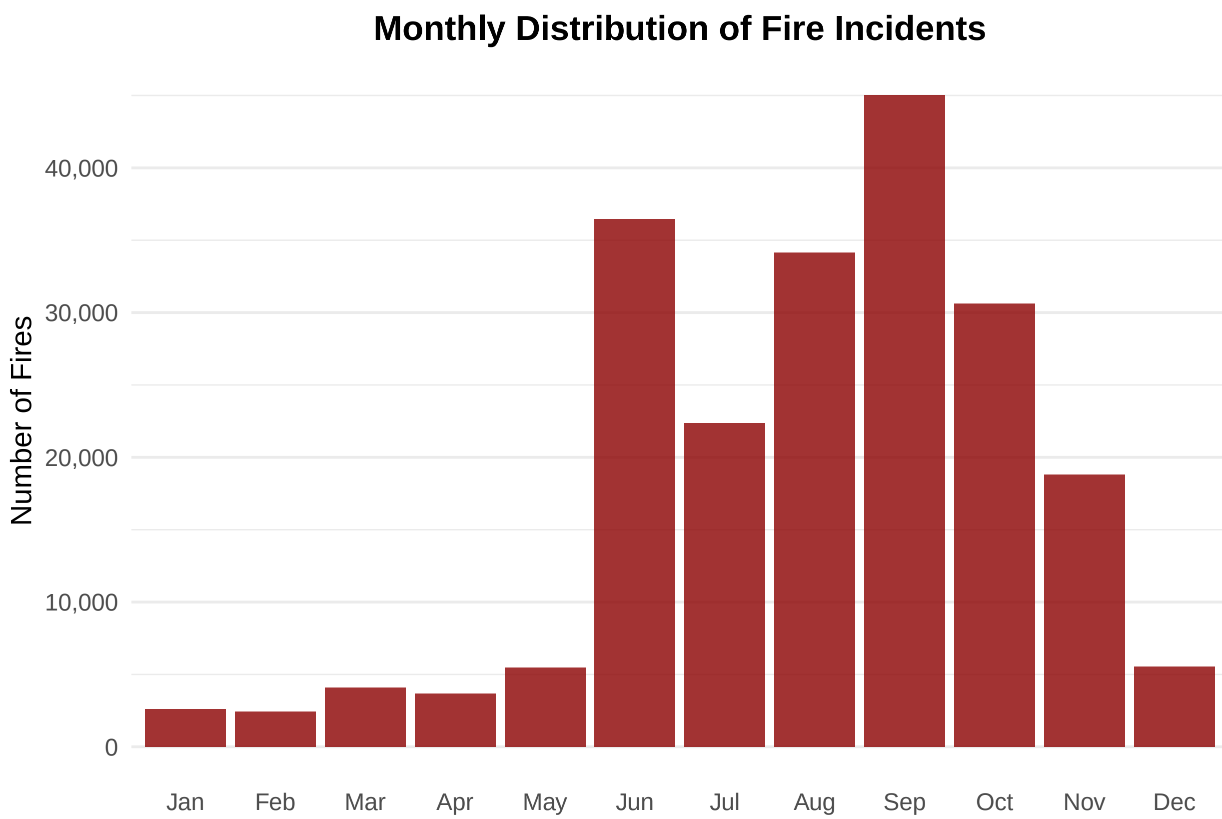


The choropleth map reveals a distinct geographical pattern. The provinces located along the Mediterranean and Aegean coasts (such as Antalya, Muğla, İzmir, and Mersin) are depicted in the darkest shades of red, indicating the highest number of fire incidents. This distribution correlates strongly with the hot-summer Mediterranean climate, where dry summers create ideal conditions for wildfires. Inner Anatolia and the Black Sea regions show relatively lower fire frequencies.

#2. Seasonal Analysis Understanding "when" fires occur is crucial for preparedness. I analyzed the monthly distribution of fire incidents to identify the peak fire season.

```
monthly_stats <- fire_joined %>%
  group_by(month) %>%
  summarise(count = n())


ggplot(monthly_stats, aes(x = month, y = count)) +
  geom_col(fill = "darkred", alpha = 0.8) +
  scale_y_continuous(labels = comma) +
  labs(title = "Monthly Distribution of Fire Incidents",
       x = "", y = "Number of Fires") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
        panel.grid.major.x = element_blank())
```



The bar chart demonstrates a clear seasonal cycle. Fire activity is minimal during winter and spring months but begins to rise sharply in June. The frequency reaches its peak during August and September.

Note: While July and August are the hottest months, the high numbers in September and October may also include agricultural stubble burning (anız yakma), which satellite sensors detect as thermal anomalies.

#3. Annual Trend and Anomalies Finally, I examined the yearly evolution of fire frequencies to detect any increasing trends or specific outlier years over the two-decade period.

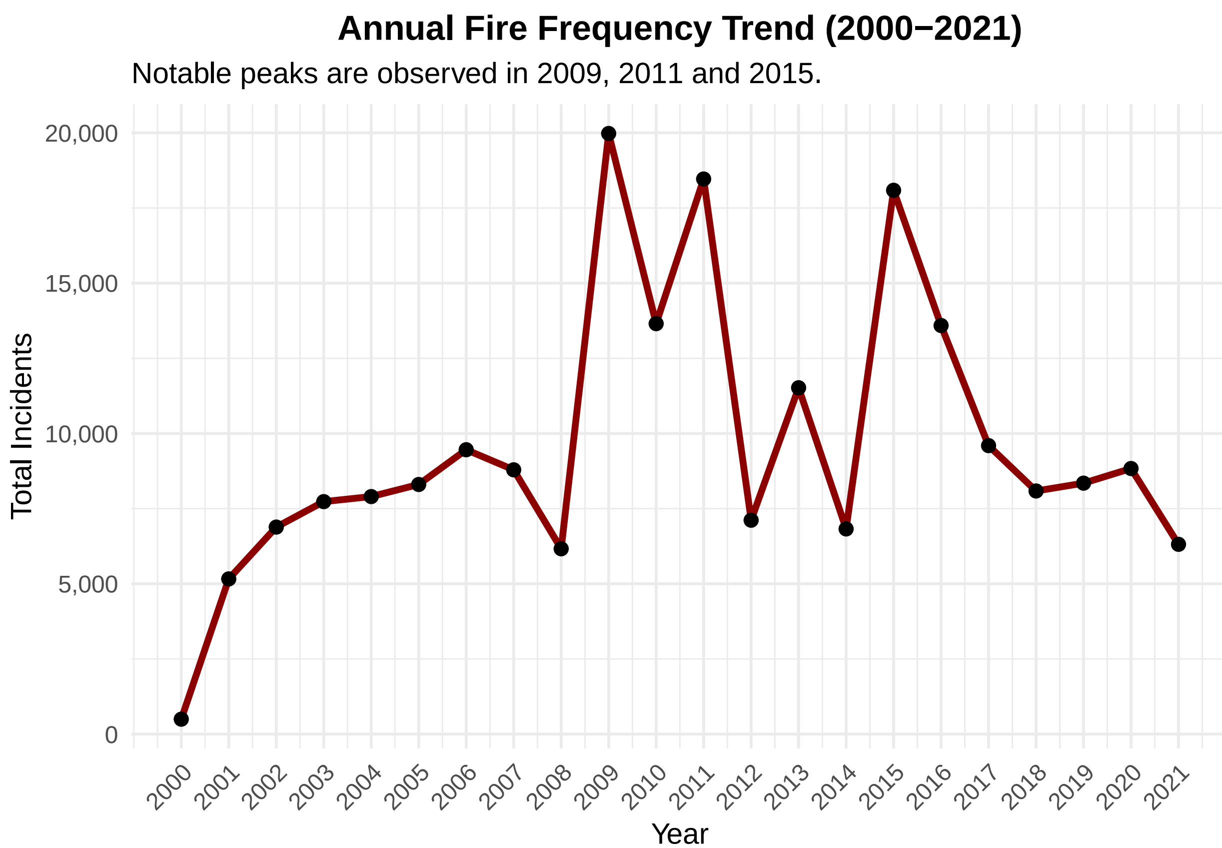
```
yearly_stats <- fire_joined %>%
  group_by(year) %>%
  summarise(count = n())


ggplot(yearly_stats, aes(x = year, y = count)) +
  geom_line(color = "darkred", size = 1.2) +
  geom_point(color = "black", size = 2) +


  scale_x_continuous(breaks = 2000:2021) +

  scale_y_continuous(labels = comma) +
  labs(title = "Annual Fire Frequency Trend (2000-2021)",
       subtitle = "Notable peaks are observed in 2009, 2011 and 2015.",
       x = "Year", y = "Total Incidents") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
```

```
    axis.text.x = element_text(angle = 45, hjust = 1))
```



The time-series analysis reveals a fluctuating trend rather than a linear increase. The data indicates that 2009 was the year with the highest number of detected fire points within the studied period. Following a decline after 2009, the frequency remained relatively stable until a significant resurgence in 2021.

The high numbers in earlier years (like 2009) might be attributed to less regulated agricultural burning practices, while the spike in 2021 represents a major recent wildfire event driven by extreme weather conditions.

Conclusion This analysis confirms that forest fires in Turkey are highly concentrated in the coastal Mediterranean climate zones and follow a strict seasonal pattern peaking in late summer. Temporally, while the highest frequency of fire points was recorded in 2009, the year 2021 marks a critical recent anomaly, highlighting the continuing risk of wildfires in the region.